Hallucination Analysis in RAG and GraphRAG Systems

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https://github.com/Mflavien01/GraphRAG-Hallucination-Analysis

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01

Introduction & Background

hallucinations · research question · the two paradigms · datasets

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RAG was meant to reduce hallucinations, yet they persist

Retrieval-Augmented Generation grounds a language model by injecting retrieved passages into the prompt.

In practice, RAG systems still:

- invent entities that are not in the context
- infer unsupported relations between real entities
- fall back on parametric memory and world knowledge

A growing body of work points at the retrieval mechanism, not only the generator, as a decisive factor.

Working definition

A hallucination is any statement in the answer that is not entailed by the retrieved evidence.

Our goal

Not to eliminate hallucinations, but to characterise WHEN, WHY and HOW they arise in each paradigm.

What is a hallucination?

Any statement in the answer that is NOT entailed by the retrieved evidence.

It is about grounding in the provided context, not real-world truth: a factually correct but unsupported answer still counts.

Invented entities
entities in the answer, absent from the retrieved context
e.g. the answer adds a co-author who is nowhere in the context

Incorrect relations
real entities linked by an unsupported relation
e.g. says an actor “directed” a film he only starred in

Unsupported facts
claims drawn from the model's own parametric memory
e.g. answers “Joe Biden” when the context names no leader

Ontology violations
subject / object types incompatible with the predicate
e.g. gives a country's “capital” as a person, not a city

One question, one hypothesis

How do the retrieval mechanisms of RAG and GraphRAG influence the frequency, nature and typology of the hallucinations an LLM produces?

Hypothesis: the two paradigms should fail differently.

RAG

Vulnerable to lexical-semantic confusion: retrieves chunks that are topically related but factually irrelevant.

GraphRAG

Vulnerable to ontological gaps, incomplete sub-graphs, and reasoning chains spanning several triples.

Objectives

Not to eliminate hallucinations, but to characterise WHEN, WHY and HOW they arise in each retrieval paradigm.

TASK 1

Question generation

547 validated QA pairs
200 text (T5) + 347 graph (LLM)
single-hop & multi-hop

TASK 2

QA pipelines

6 pipelines, one generator
retrieval is the only variable

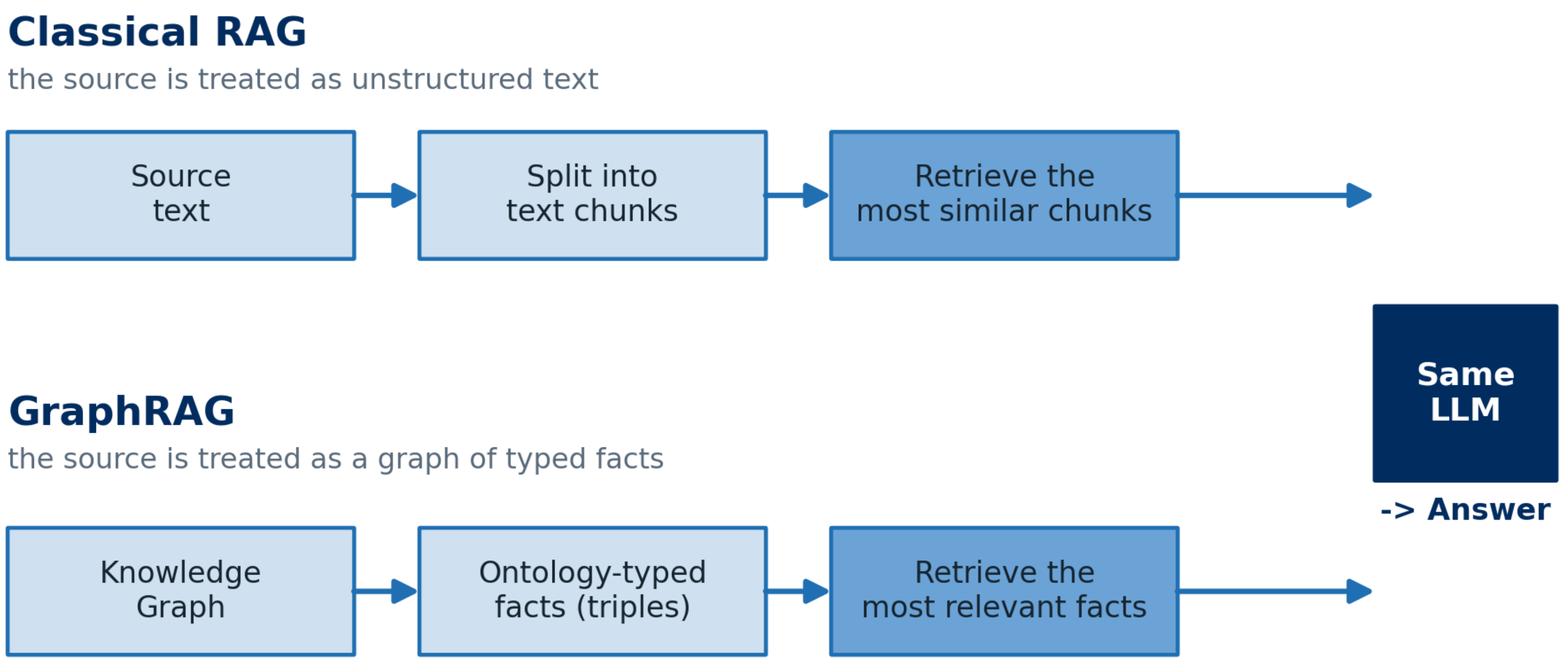
TASK 3

Hallucination evaluation

MIRAGE + FactCC + SBERT
quantitative + qualitative

Three sequential tasks — each feeds the next

RAG vs GraphRAG: two ways to retrieve



Same generator. The only difference is WHAT is retrieved: free-form text vs. structured, typed facts.

Same generator either way. We detail our own RAG and GraphRAG, with the exact technologies, in Task 2.

02

Task 1 · Question Generation

building 547 balanced, validated QA pairs

Two complementary knowledge-graph benchmarks

Text2KGBench-LettrIA

- 19 domain ontologies, strict typing
- 4,860 annotated sentences (2,012 dev)
- ~14,000 high-fidelity triples
- Object vs datatype properties
- JSONL per category

OSKGC

- 10,183 entries, 19 / 57 sub-categories
- 207 entity types, 382 relations
- subsumption hierarchy of depth 4
- schema placed at the centre of the task
- XML per sub-category

Why two? LettrIA is denser but flatter; OSKGC is finer-grained and deeper. Together they broaden coverage and guard against single-benchmark artefacts.

A balanced, dual-provenance question set

From TEXT · fine-tuned T5

valhalla/t5-small-qa-qg-hl: answer extraction, then answer-aware question generation. Spans must appear literally in the sentence.

From GRAPH · strictly-prompted GPT-4o

A deterministic graph-walker: strict negative grounding rules, explicit A -[rel]-> B chains, and a multi-hop feasibility check so it never fabricates a fake chain.

Unfolded into a set sliced three ways: paradigm × provenance × complexity.

547
validated QA pairs

200
text-derived (T5)

199
single-hop (graph)

148
multi-hop (graph)

We benchmarked models and prompts before generating

Model benchmark

Five candidates scored on five criteria (grammar, faithfulness, fluency, relation visibility, chain coherence):

GPT-4o · Claude Sonnet 4.6 · Gemini 2.5 Flash · Mistral 7B · Llama 3.1 8B

→ **GPT-4o selected for chain traceability.**

Prompt optimisation

A: zero-shot baseline (no format)

B: one-shot with explicit A -[rel]-> B chain

C: zero-shot strict JSON, weaker chains

→ **Variant B: the chain is an audit trail for hallucination traceability.**

Then: every QA pair manually reviewed. ~30% of T5 questions reformulated into compositional, multi-hop variants to restore the complexity balance.

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Task 2 · QA Pipelines

six pipelines, retrieval as the only variable

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Six pipelines, one generator: retrieval is the only variable

Shared by all: generator Qwen3-30B-A3B (MoE, EURECOM gateway) · same source corpora · same 547 questions.

BASELINES

RAG
FAISS L2 over text chunks

GraphRAG
cosine over verbalised triples

+ CROSS-ENCODER HYBRIDS

RAG + CE
hybrid recall, reranked

GraphRAG + CE
hybrid triples, reranked

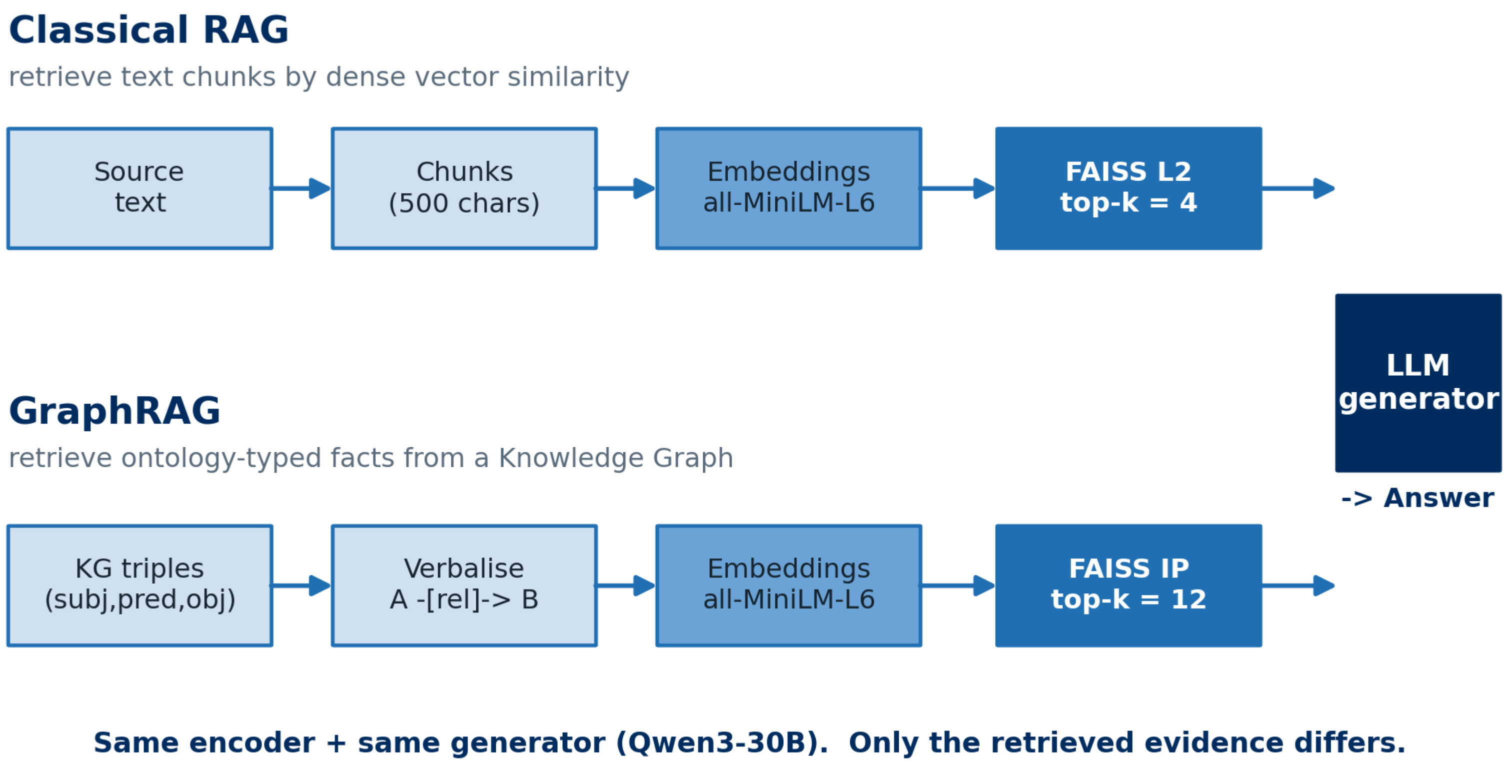
Graph-Chunk + CE
dual retriever, LLM fuses

ABLATION

RAG Parent-Child
small child → large parent

An earlier RRF fusion is reported as a measured intermediate step, not the final design.

Our RAG & GraphRAG baselines, in detail



Shared encoder (all-MiniLM-L6-v2) and generator (Qwen3-30B). Only the index differs: text chunks (FAISS L2, k=4) vs. verbalised triples (FAISS IP, k=12).

How our GraphRAG retriever evolved

v1 — Entity linking + graph walk

Anchor node via fuzzy string match (RapidFuzz)
then BFS traversal of the sub-graph

X Leaf node -> no outgoing edges -> empty context -> the model hallucinates

supervisor feedback

v2 — Statement-level retrieval

Verbalise each triple A -[rel]-> B
embed + cosine similarity (FAISS IP, k=12)

OK Every triple is a candidate, regardless of position -> leaf-node failure solved

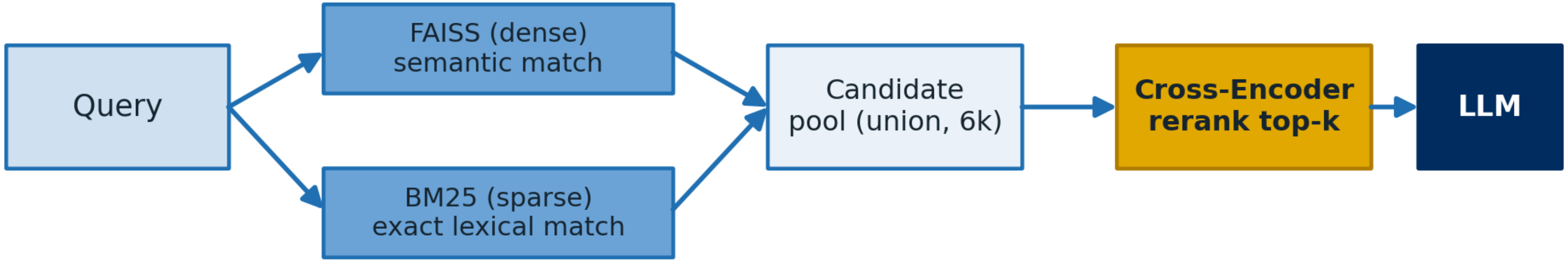
Residual limitation

Cosine over individual triples cannot recover a
two-hop reasoning chain in a single pass -> motivates the hybrids

Recall widely, then rerank precisely

Recall -> Rerank

Stage 1 widens recall (dense + sparse); Stage 2 reads each (query, passage) pair jointly.

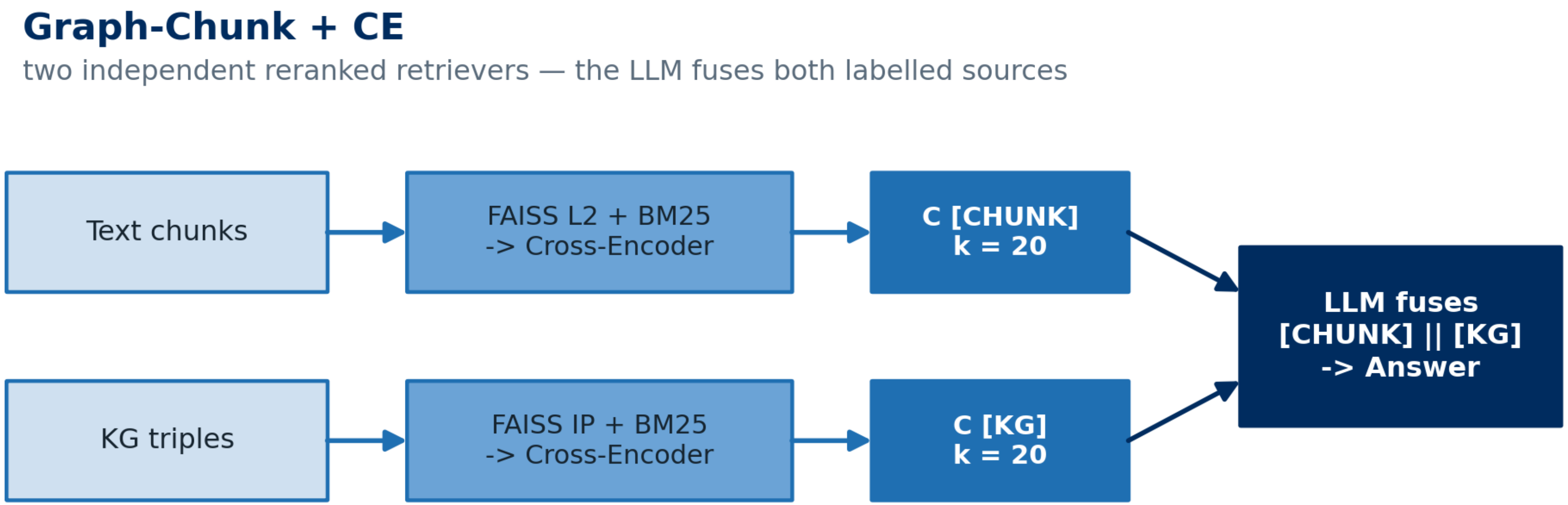


A measured iteration: RRF -> Cross-Encoder

RRF (rank-only fusion) was content-blind -> faithfulness stagnated.

On Graph-Chunk: 0.533 @ 13.5% -> 0.540 @ 11.0% after switching to the cross-encoder.

Graph-Chunk + CE: fusing both worlds



Lexical coverage (chunks) + relational structure (triples), always both present

-> lowest abstention of all pipelines (10.4%) · recovers the multi-hop gap

Ablation: parent-child chunking

Parent chunk (1500 chars) → injected as context

Child 1
150 chars

Child 2 ✓
matched

Child 3
150 chars

child matched → inject whole parent into prompt

Result: neutral-to-negative

Faithfulness 0.534 vs 0.547 (flat RAG)

Abstention 15.2% vs 17.2%

SBERT 0.445 vs 0.446

Takeaway

Wider context dilutes single-sentence answers (T5: 0.605 vs 0.636).

What cuts hallucination = reranked, filtered evidence, not more text.

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Task 3 · Evaluation & Results

MIRAGE · FactCC · Sentence-BERT

How we measured hallucination

MIRAGE + FactCC

A BERT classifier judges whether the answer is supported by the source sentence.

faithfulness = 1 − score

We score on answered entries only; a hallucination is faithfulness < 0.5.

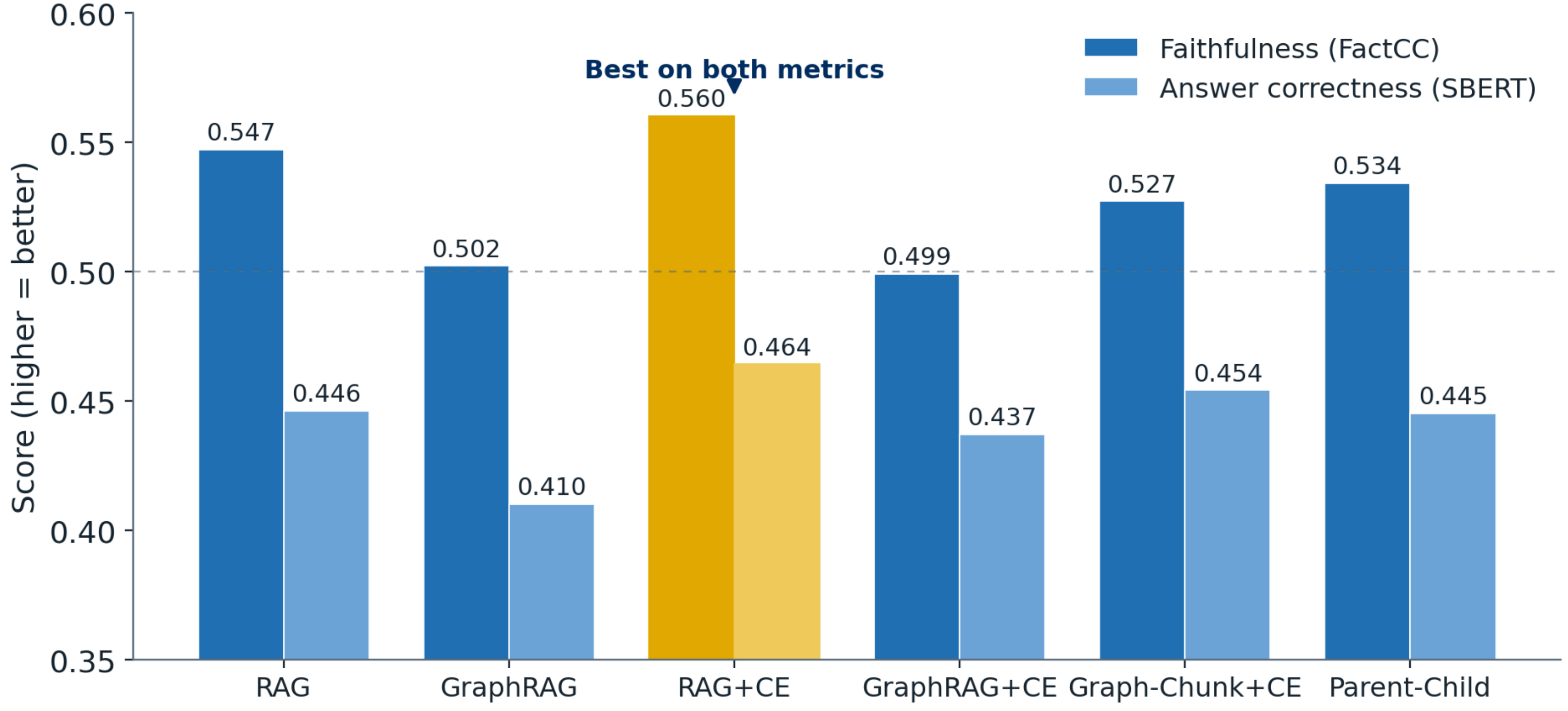
SBERT correctness

Cosine similarity to the ground-truth answer: source-independent sanity check.

Coverage errors are separated from hallucinations.

An abstention ("the context does not contain…") is detected by a regex on the answer, not by the FactCC score, and excluded from faithfulness. This keeps GraphRAG's leaf-node abstentions from inflating its hallucination rate.

RAG + CE leads on both metrics



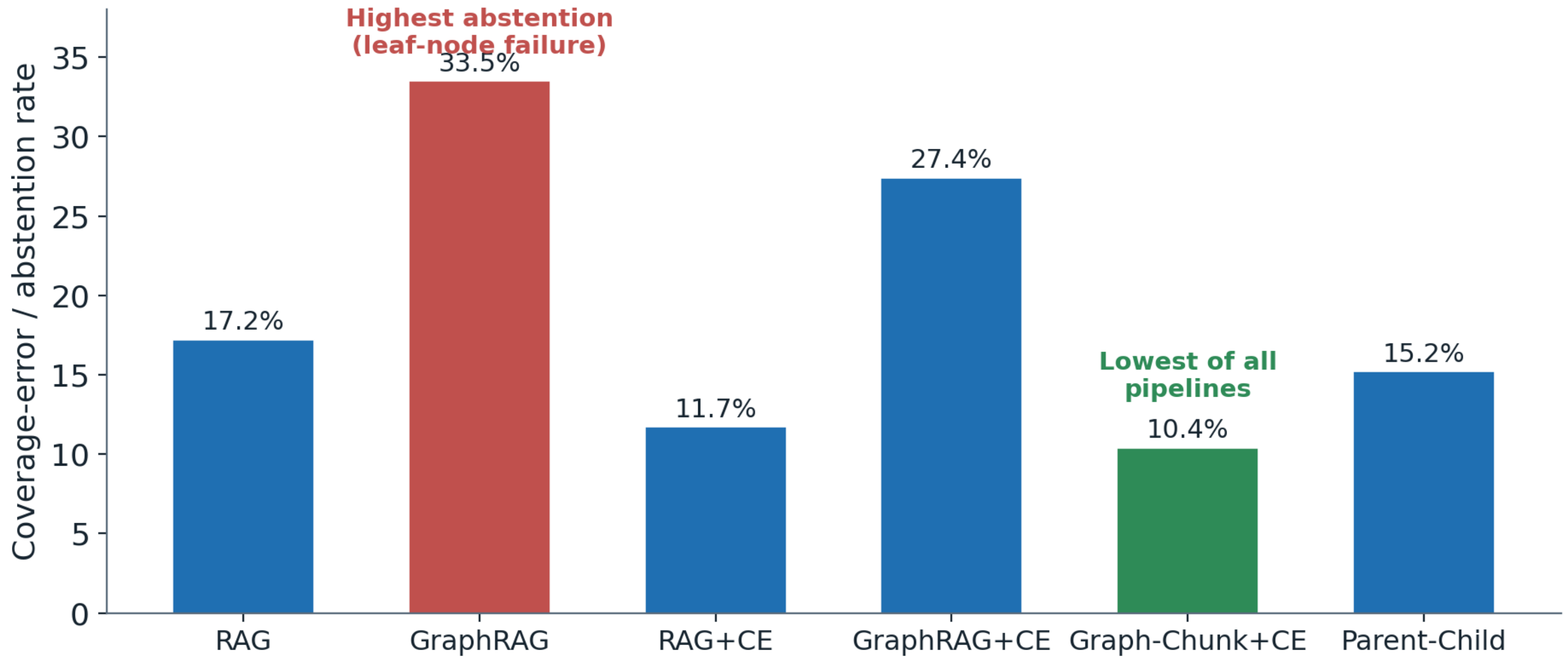
Takeaway

Both metrics agree on the top.

RAG+CE has the highest mean (0.560 / 0.464), but its gain over RAG is within noise.

The robust gap is text vs GraphRAG (see significance).

Where the graph approach wins: abstention



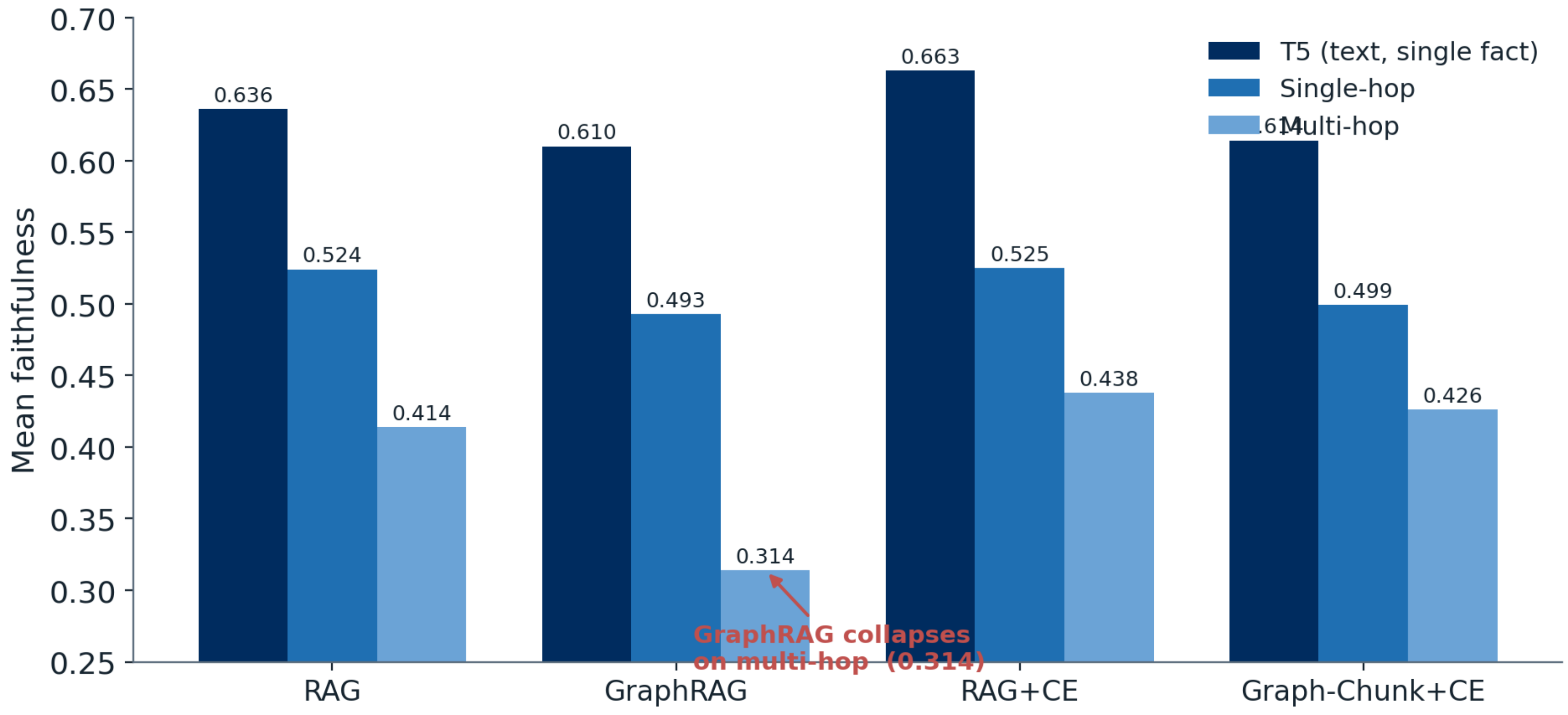
Takeaway

GraphRAG abstains most: 33.5%, driven by leaf nodes.

Graph-Chunk+CE abstains least of all: 10.4%.

Always having a text view stops the empty-context failure.

Multi-hop is where the paradigms diverge



Takeaway

GraphRAG collapses on multi-hop: 0.314, ~10 pts below RAG.

Both hybrids recover it: RAG+CE 0.438, Graph-Chunk+CE 0.426.

A single reranked chunk can carry a whole chain.

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Are the gaps significant?

Paired bootstrap + permutation test on per-question faithfulness (questions answered by both):

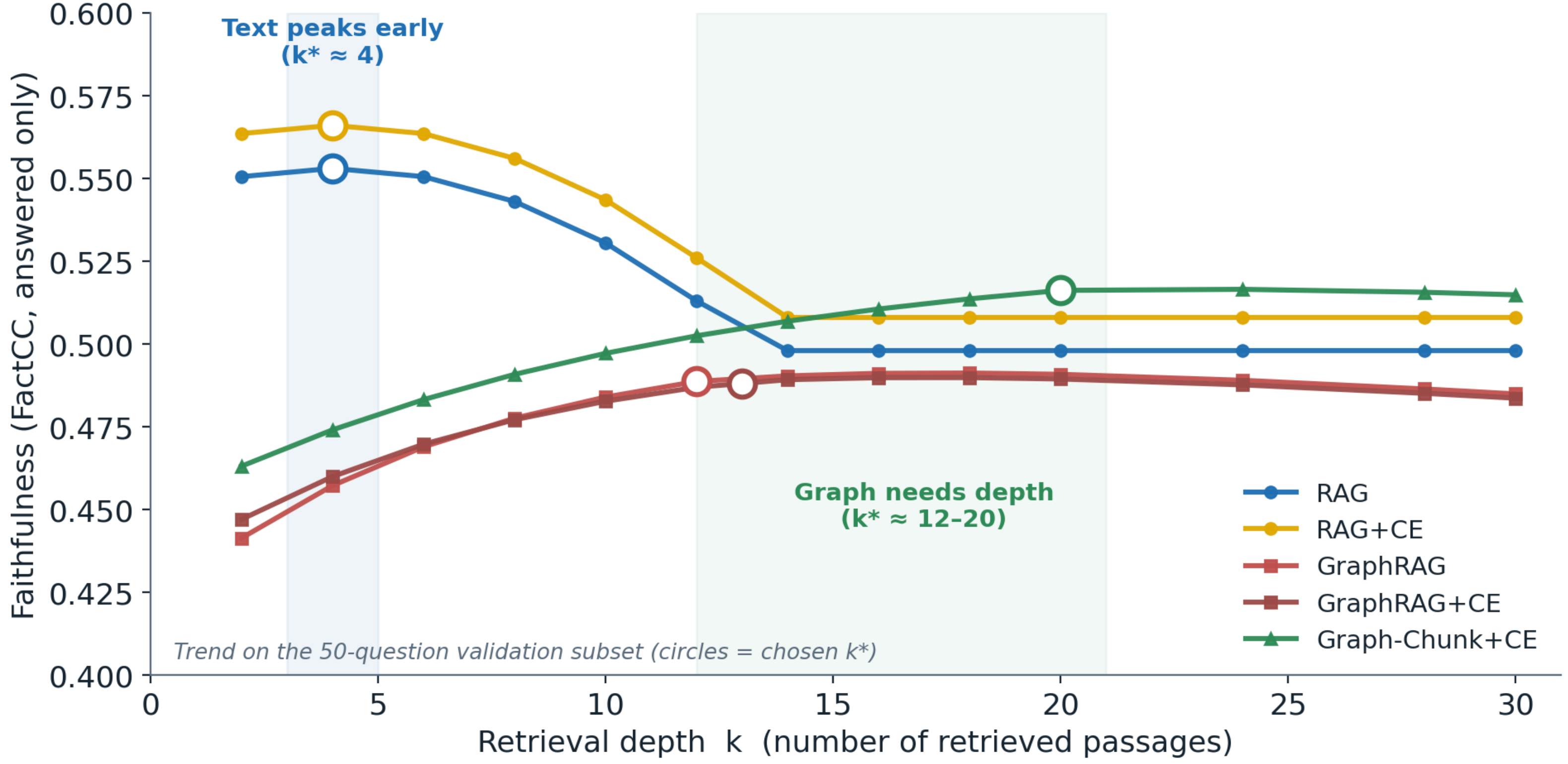
RAG + CE vs RAG	**+0.012**	n.s. ($p = 0.40$)
RAG vs GraphRAG	**+0.055**	significant ($p < 0.01$)
GraphRAG + CE vs GraphRAG	**-0.007**	n.s. ($p = 0.65$)
RAG vs GraphRAG (graph-only Q)	**+0.055**	significant ($p = 0.04$)

Takeaway

The two best pipelines (RAG+CE and RAG) are statistically tied on faithfulness.

The robust, significant gap is text vs GraphRAG, and it holds even on graph-derived questions.

Text saturates early, graph needs depth



Takeaway

Text peaks at k≈4, then extra chunks dilute ('lost in the middle').

Graph climbs to k≈12–20: atomic triples need volume, and depth also cuts abstention.

The ranking is stable across k, not an artefact of one choice.

Four failure modes, mapped to their causes

Parametric memory leak
RAG

Model answers from pre-trained knowledge, not the retrieved context

Surface-form mismatch
GraphRAG

Triple labels differ from the expected wording -> scorer penalises

Multi-hop context gap
GraphRAG

Single-pass triple retrieval misses the intermediate hop of a chain

Leaf-node abstention
GraphRAG

Entity with no edges -> empty context -> honest 'I cannot answer'

From a-priori typology to observed failures.

We started from four textbook categories (invented entities, wrong relations, unsupported facts, ontology violations).

Genuine ontology violations were rare: graph answers come from typed triples, so they seldom break the schema.

The real failures cluster in these four, one per root cause.

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Conclusion

findings · contributions · limitations

Three lessons from the project

1

Retrieval is a first-class variable.

The same generator, given different context, swings from faithful to hallucinating. Retrieval shapes failure.

2

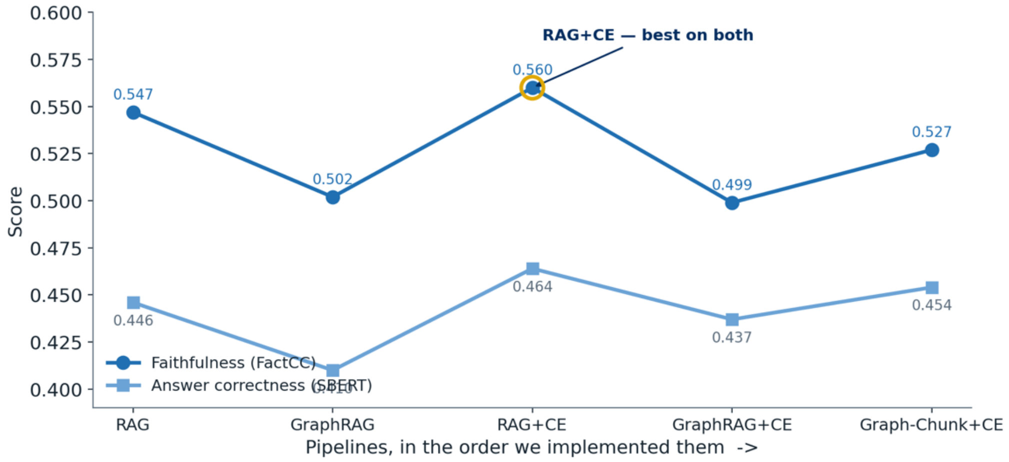
The same component plays different roles.

Cross-encoder reranking lifts the text pipeline on every axis, but on triples it only cuts abstention.

3

Applying metrics correctly is consequential.

The FactCC sign-inversion, FactAcc's unsuitability, and the k-sweep were findings in their own right.



Performance trajectory, in implementation order.

What we delivered

Evaluation dataset
547 validated QA pairs over two KG benchmarks, dual provenance, single- and multi-hop.

Model + prompt studies
Five-LLM benchmark and a three-variant prompt study for graph question generation.

Six reproducible pipelines
Two baselines, three cross-encoder hybrids incl. the novel Graph-Chunk+CE, one ablation.

Combined evaluation
MIRAGE + FactCC + SBERT, plus a four-category qualitative failure taxonomy.

Methodology analysis
FactCC sign-fix, FactAcc rejection, surface-form bias, and the retrieval-depth study.

Open, reproducible code
Full pipeline, checkpoints and retries on GitHub.

Limitations and next steps

Limitations

- FactCC was trained on news summarisation and scores surface form.
- SBERT is depressed by the verbosity asymmetry (sentences vs short spans).
- A single generator (Qwen3-30B): generator vs retrieval effects are entangled.
- Mixed-provenance questions create an alignment bias that favours text.

Future work

- A dedicated abstention classifier for a clean faithful / hallucinated / abstained split.
- A provenance-controlled comparison on KG-only questions, GraphRAG's native setting.
- Iterative retrieval with sub-question decomposition (IRCoT, ReAct) for multi-hop.

Conclusion

The retrieval mechanism determines not just how often a model hallucinates, but how it fails.

RAG + CE

Highest mean faithfulness (0.560) and correctness (0.464); its gain over RAG is within noise.

Graph-Chunk + CE

Strongest graph variant: lowest abstention (10.4%), multi-hop gap recovered.

Thank you. We would be glad to take your questions.

https://github.com/Mflavien01/GraphRAG-Hallucination-Analysis

Appendix · Backup material

Detailed tables, examples and methodology, kept on hand for the discussion.

All pipelines, all metrics

Pipeline	Answered	Coverage err.	Faithfulness	Hallucination	SBERT
RAG	453 / 547	17.2%	0.547	44.6%	0.446
GraphRAG	364 / 547	33.5%	0.502	49.2%	0.410
RAG + CE	**483 / 547**	**11.7%**	**0.560**	**43.9%**	**0.464**
GraphRAG + CE	397 / 547	27.4%	0.499	49.4%	0.437
Graph-Chunk + CE	**490 / 547**	**10.4%**	**0.527**	**47.1%**	**0.454**
RAG Parent-Child	464 / 547	15.2%	0.534	46.1%	0.445

Faithfulness and hallucination are computed on answered entries only. Gold = best overall (RAG+CE); green = lowest abstention (Graph-Chunk+CE).

Breakdown by question type

FactCC faithfulness

Pipeline	T5	Single	Multi
RAG	0.636	0.524	0.414
GraphRAG	0.610	0.493	0.314
RAG + CE	**0.663**	**0.525**	**0.438**
Graph-Chunk + CE	0.614	0.499	0.426

SBERT correctness

Pipeline	Global	T5	Single	Multi
RAG	0.446	0.479	0.436	0.402
GraphRAG	0.410	0.435	0.403	0.379
RAG + CE	**0.464**	**0.504**	**0.449**	**0.418**
Graph-Chunk + CE	0.454	0.492	0.441	0.410

Multi-hop is the divergence point: GraphRAG drops to 0.314 (~10 pts below RAG); both hybrids recover it. The same ordering holds on the source-independent SBERT metric.

Answer type & the parent-child ablation

Numeric vs textual answers

Pipeline	Numeric	Textual	Δ
RAG	0.620	0.537	+0.083
GraphRAG	0.597	0.491	+0.106
GraphRAG + CE	0.556	0.492	+0.064

Why?

Numeric values are stated verbatim in chunks and as triple objects, so both retrievers reproduce them more reliably than free-form text.

Parent-child vs baselines

Pipeline	Answ.	Cov.	Faith.	SBERT
RAG baseline	453	17.2%	0.547	0.446
Graph-Chunk + CE	490	10.4%	0.527	0.454
RAG Parent-Child	464	15.2%	0.534	0.445

Why parent-child is neutral-to-negative

Larger 1500-char parents dilute single-sentence (T5) answers (0.636 → 0.605). What cuts hallucination is reranked, filtered evidence, not just more text.

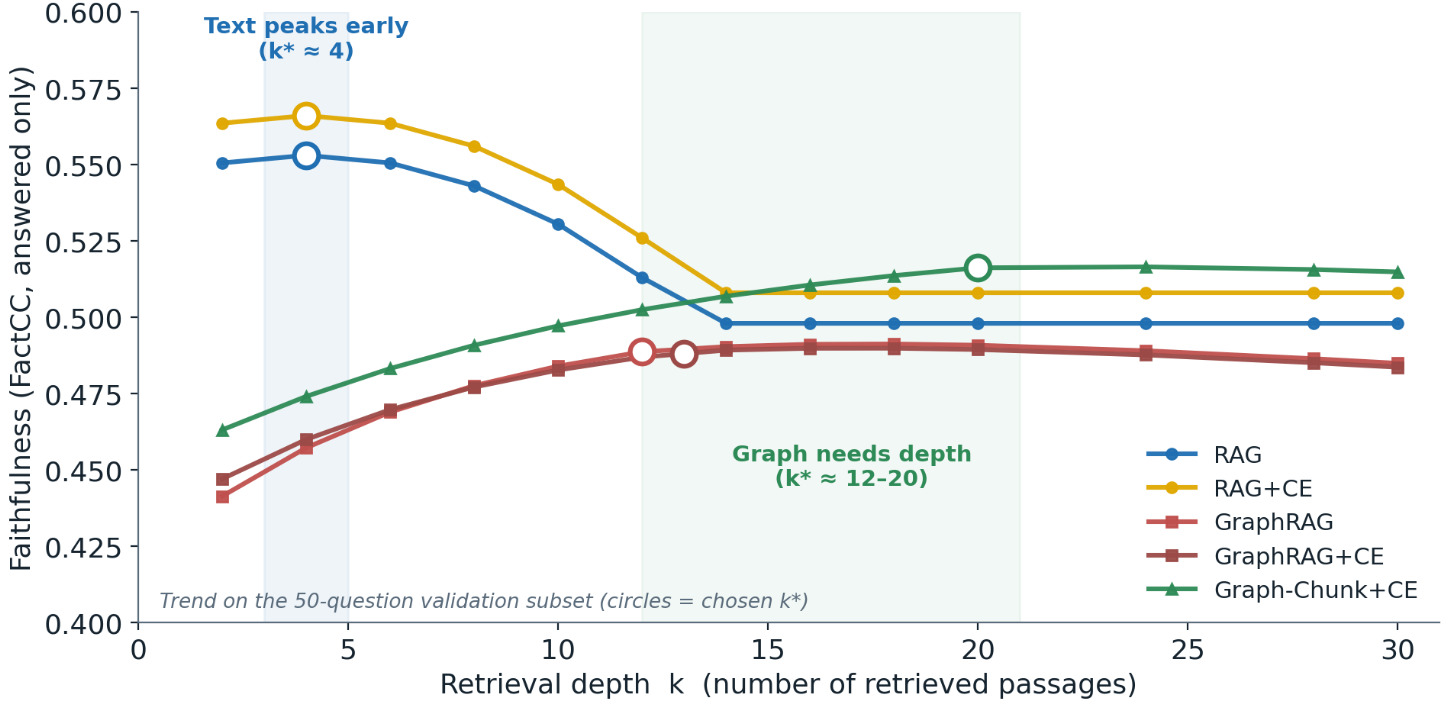
Configurations & technical stack

Pipeline	Retriever	Reranker	k
RAG	FAISS L2 (chunks)	—	4
GraphRAG	FAISS IP (triples)	—	12
RAG + CE	FAISS L2 ∪ BM25	Cross-Encoder	4
GraphRAG + CE	FAISS IP ∪ BM25	Cross-Encoder	13
Graph-Chunk + CE	both hybrid legs	Cross-Encoder	20+20
RAG Parent-Child	FAISS L2 (child→parent)	—	4

Stack

- Generator: Qwen3-30B-A3B (MoE, 3B active), EURECOM LiteLLM gateway
- Encoder: all-MiniLM-L6-v2
- Sparse: BM25, top-6k pool, union with FAISS
- Reranker: ms-marco-MiniLM-L-6-v2
- Chunking: 500 / 50 overlap; parent 1500 / child 150
- Orchestration: checkpoint-resume + exponential retry (≤300 s)

Retrieval-depth (k) study



Optimal k (frozen on validation)

RAG 4 · RAG+CE 4 · GraphRAG 12 · GraphRAG+CE 13 · Graph-Chunk+CE 20 per leg

RRF → Cross-Encoder

Measured on Graph-Chunk: 0.533 @ 13.5% (RRF) → 0.540 @ 11.0% (CE). Protocol: coarse grid {2,5,10,15,20}, then granular to 30; frozen before the test set.

Evaluation methodology in detail

FactCC (via MIRAGE)

- BERT classifier, fine-tuned on CNN/DailyMail summarisation (domain mismatch with short QA).
- MIRAGE exports P(incorrect); we use faithfulness = 1 − score.
- Hallucination = answered entry with faithfulness < 0.5.
- Coverage errors detected by regex on the answer ("context does not contain"…), excluded from faithfulness.

Cross-checks & a rejected metric

- SBERT cosine (answer ↔ gold), source-independent; low absolute range 0.39–0.49 (verbosity asymmetry).
- Pearson r(faithfulness, SBERT) = +0.24, vs −0.24 before the orientation fix.
- FactAcc (REBEL) rejected: needs full sentences; on 1–3 word answers it self-loops (mean 0.055, 5/6 at zero).
- FactCC surface-form bias documented against ontology labels.

Concrete hallucination examples

Parametric memory leakage · RAG

Leadership question whose retrieved chunks never name the leader → model answers "Joe Biden" from memory. Maybe true, but not grounded.

Surface-form mismatch · GraphRAG

Dentcheva -[field]-> mathematical-optimisation; gold says "Mathematical optimization". Correct, but the scorer penalises the surface difference.

Multi-hop context gap · GraphRAG

"Where did the person who discovered Camilla die?" needs Pogson → Chennai. Single-pass triple retrieval grabs only one hop.

Leaf-node abstention · GraphRAG

Target entity has no outgoing edges → empty context → honest "the context does not contain information about X" (a coverage error).

Question generation: method detail

Text · fine-tuned T5 (two steps)

- Step 1, answer extraction: keep only spans appearing literally in the sentence.
- Step 2, question generation: answer wrapped in <hl> markers.
- 60M-param model: tends to ask trivial single-fact questions.
- → ~30% manually reformulated into compositional multi-hop variants.

Graph · strictly-prompted GPT-4o (temp 0)

- Variants A / B / C tested → B chosen (one-shot, explicit chain).
- Strict negative rules: "do NOT invent", "do NOT use world knowledge".
- Multi-hop feasibility check: connected if triples share an entity; else a placeholder, never a fake chain.
- Fixed JSON schema + retry layer for parsing / API errors.

Every one of the 547 pairs was manually reviewed for fluency, grounding and chain validity.

Datasets: full comparison

	Text2KGBench-LettrIA	OSKGC
Source corpus	DBpedia–WebNLG	WebNLG
Categories	19	19 / 57 (by triple count)
Annotated sentences	4,860 (2,012 dev)	10,183
Ontology	hierarchical, strict typing	hierarchical, max depth 4
Properties	object vs datatype	single relation type
Storage	JSONL per category	XML per sub-category
Dedicated metric	hierarchical precision/recall	Structural Similarity

Proportional sampling

Categories are heavily imbalanced, e.g. city 10.8% vs monument 0.9%.

Stratified sampler: proportional share, floor of 1 per category, fixed seed, shuffled. n = 100 per dataset.

Selected references

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